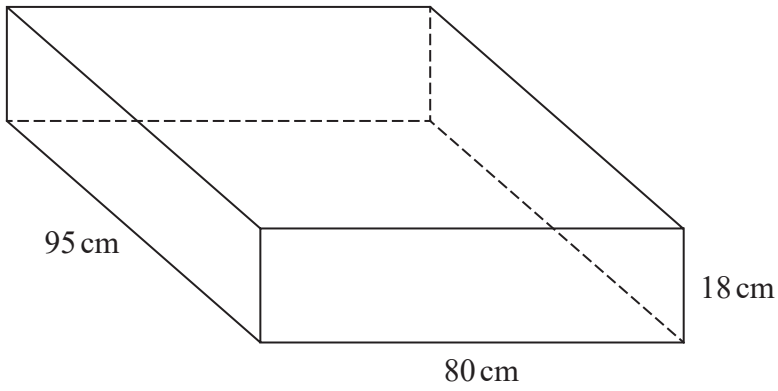
3 A sofa has 6 identical cushions.
Each cushion is a cuboid 18 cm by 80 cm by 95 cm.



The cushions are covered with a protective spray.
The protective spray is in cans.

The label on each can has this information.

Spray in this can covers 4 m^2

(a) Work out how many cans are needed to cover the 6 cushions with protective spray.

80cm = 0.80m; 95cm = 0.95m; 18cm = 0.18m.

1 mark
Convert lengths to m

Three different face areas (m^2):

0.80×0.18=0.144; 0.95×0.18=0.171; 0.80×0.95=0.760.

1 mark
At least two faces

One cushion: 2(0.144 + 0.171 + 0.760) = 2.15 m^2.

1 mark
All six faces

Six cushions: 6 × 2.15 = 12.9 m^2.

1 mark
Six cushions ÷4

Cans: 12.9 ÷ 4 = 3.225, so buy 4 cans.

1 mark
Round up:4 cans

(5)

The information on each label is inaccurate.
The spray in each can covers 10% more than 4 m^2.

(b) How will this affect the number of cans needed for the 6 cushions?
You must show how you get your answer.

New coverage = 4 × 1.10 = 4.4 m^2 per can.

1 mark
New coverage4.4

12.9 ÷ 4.4 = 2.9318... , so buy 3 cans.

Yes: the number reduces from 4 to 3.

1 mark
One fewer can

(2)

(Total for Question 3 is 7 marks)